BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 10231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The relationship between rhizobium bacteria and leguminous plants is an example of .....

(A) parasitism
(B) mutualism
(C) commensalism
(D) none of these

Q2. The science of fermentation is known as .....

(A) cytology
(B) histology
(C) genetics
(D) zymology

Q3. Any substance that is used to modify the physiological systems for the benefit of the recipient is .....

(A) medicine
(B) drugs
(C) vaccine
(D) narcotics

Q4. Gouty arthritis is caused due to higher level of ..... in blood.

(A) urea
(B) ammonia
(C) uric acid
(D) none of these

Q5. The new root develops from .....

(A) radicle
(B) plumule
(C) micropyle
(D) hilum

Q6. Blood group is an example of .....

(A) continuous variation (B) discontinuous variation

(C) gene flow (D) natural selection

Q7. In human, the main organ which works for homeostasis is .....

(A) lungs (B) skin

(C) kidneys (D) all of these

Q8. Insulin is a hormone and is secreted by .....

(A) liver (B) pancreas

(C) stomach (D) both A&B

Q9. Which one is the smallest bone of the human body?

(A) stapes (B) incus

(C) malleus (D) hyoid

Q10. High fever, cold and cough with sputum production are symptoms of .....

(A) emphysema (B) asthma

(C) pneumonia (D) bronchitis

Q11. The gaseous exchange in leaves and young stems occur through .....

(A) stomata (B) air spaces

(C) cuticle (D) all of these

Q12. Waste products excreted by the kidneys are .....

(A) salts, water and CO_2 (B) urea, water and salts

(C) urea and water (D) urea and salts

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Write down the causes, symptoms and treatment of bronchitis.

(ii) How do kidneys control the composition of blood?

(iii) Define neuron. Draw the structure of a typical neuron.

(iv) What is the difference between short sightedness and long sightedness?

(v) Differentiate between elastic cartilage and hyaline cartilage.

(vi) Enlist the bones of forelimbs.

(vii) Name the four whorls present in a flower and also identify components of each whorl.

(viii) What is crossing over? How can crossing over bring variations?

(ix) Define abiotic components of ecosystem. Write down the names of abiotic components of ecosystem.

(x) Write down the names of the products formed through fermentation.

(xi) Enlist the bad effects of caffeine and cocaine.

SECTION-C

Maximum Marks: 21

Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Describe inspiration and expiration.

(ii) Discuss lungs as homeostatic organs.

Q3.

(i) What is germination? Discuss the necessary conditions for germination.

(ii) Draw a cross between two pea plants. One of them has round yellow seeds (RRYY) while other has wrinkled green seed (rryy).

Q4.

(i) Discuss pyramid of biomass.

(ii) Write down objectives of genetic engineering.

Q5.

(i) Distinguish between herbivores, carnivore and omnivores.

(ii) Write down behavioural symptoms and psychological symptoms of drug abuse/addiction.